AfyaSasa Pharmacy — implementation, verification and sign-off

Date: 22 October 2026

Tenant: `jalaram`

Schema: `demo`

API: `/api/v1`

HEAD: `247d026` (`Blood bank and stock`) on `main`

Release boundary: local only. Nothing was committed, pushed, merged or deployed.

1. Executive summary and sign-off status

The Pharmacy module now runs as one workflow on the existing inventory ledger, charges and payments. Prescriptions and walk-in POS prepare stock as **holds** (reservations), cashier clearance uses existing `charges` / `payment_transactions`, and issue posts `DISPENSE`. Stock Oversight is available to `inventory:read` without `settings:manage`. A real stock-take session (baseline → count → submit → reject/recount → independent approve → one-time ledger post) is in the local app.

Sign-off: not ready for controlled hospital release.

Ready for **staff-in-the-loop local verification** after an authorised user signs in with their live password. Not ready to treat as released until the 15 browser scenarios are executed with that login, a second `inventory:manage` user posts a variance, and isolated migration tests run on a disposable database (not this compose volume).

Demo login `it@jalaram.co.ke` / `ChangeMe123!` still returns **Authentication is required** (AUTH_REQUIRED). Passwords were not reset.

2. Original problems and scope

Pharmacy, inventory, cashier and Hospital Control Center existed as separate screens. Stock-take posted immediately from CSV. Walk-in billing required a patient. Stock Oversight was missing. Holds were not distinguished from on-hand. This work connected those pieces; it did not create a second pharmacy application.

3. Final module structure and navigation

Screen	Permission	Purpose
Pharmacy	`pharmacy:read`	Queue, prescriptions, issue
Pharmacy Sales POS	`pharmacy:read`	Multi-line OTC / walk-in cart
Inventory & Store	`inventory:read`	Receive, transfers, stock-take session + legacy sheet
Stock Oversight	`inventory:read`	HCC Overview / Stores / Control / Reports
Payments	`payments:collect` (existing)	Pharmacy sale lookup and queue

HCC configuration remains `settings:manage`. Stock Oversight does not grant it.

4. Prescription-to-dispensing workflow

1. Clinician writes a pharmacy clinical order. Stock is unchanged.

2. Pharmacy prepares (`POST /inventory/sales/prepare`). FEFO lots go to `demo.inventory_reservations` (`held`). `qty_on_hand` does not change.

3. `PaymentsService.upsertPharmacyCharge` opens/updates the bill.

4. Cashier collects via `POST /payments/manual` (or STK).

5. Issue (`POST /inventory/dispense/prescription`) runs `evaluatePharmacyClearance` inside the transaction. Unpaid / partial / insurance-without-policy fail closed.

6. Issue consumes holds, reduces on-hand, posts `DISPENSE`.

7. Cancel releases `held` rows. It does not refund collected money.

Status: Implemented but not fully verified (unit/clearance tests pass; browser issue path not executed).

5. POS-to-cashier workflow, including walk-ins

`PharmacyPosPanel` searches catalogue / SKU / barcode, shows selling price and available quantity (on-hand minus held), and prepares a multi-line cart with optional patient or walk-in name/phone.

Walk-in charges use additive nullable `patient_id` on `demo.charges` and `demo.payment_transactions`. Clinical billing still expects a patient at the application layer. No dummy patient is created.

Cashier: `GET /payments/pharmacy-queue`, `GET /payments/pharmacy-sale/:saleId`. Payment attempt key is stable in the checkout panel. Receipt uses existing print helper.

POS discounts, VAT and refunds were not invented.

Status: Implemented but not fully verified.

6. Stock reservation, batch, FEFO and ledger

- Available = on-hand − held reservations.
- Expired lots remain ineligible for receive/dispense (`isExpiredBatch`).
- Recalled / quarantined pharmacy lots: **Blocked** — no authorised movement types beyond blood-bank structures.
- Ledger of record remains `demo.inventory_transactions`. Reservations are not a second ledger.

Status: Implemented and verified at unit level (`stock-available`, `stock-posting`, `pharmacy-clearance`). Isolated `inventory-pg` not run (`inventory_it` not created).

7. Stock-take and approval workflow

Deficient path: CSV `POST /inventory/items/import` with `confirmStockTake` still posts immediately for `inventory:manage` (legacy, labelled as such).

Required path (new):

1. `POST /inventory/stock-take-sessions` snapshots on-hand lots at a location (`baseline_at`).

2. Counts are saved per line. Variance requires a reason (≥3 characters). Counting does not change stock.

3. Submit requires every line counted.

4. Reject returns the session to recount; baseline and history remain.

5. Approve/post requires a **different** `inventory:manage` (or `settings:manage`) user. Same submitter is rejected (`STOCK_TAKE_INDEPENDENT_APPROVAL`).

6. Post uses existing `adjustOnHand` / `STOCK_TAKE` ledger, session id as idempotency key, and aborts if on-hand drifted from baseline.

7. Posted sessions return the existing result on retry.

Tables: `demo.inventory_stock_take_sessions`, `demo.inventory_stock_take_lines` (additive; applied on local compose Postgres `afyasasa` / schema `demo` on 22 Oct 2026). `TYPEORM_MIGRATIONS_RUN` remains `false`.

Status: Implemented but not fully verified (rules tests pass; browser count/approve not executed).

8. Control Center configuration and reports

Overview tiles: on-hand, reserved/held, available, pending stock takes, low stock, near-expiry (90 days), expired lots.

Stores & Transfers: existing transfer/requisition APIs; no duplicated GRN.

Stock Control: pharmacy batches, expiry, history; recall/quarantine/disposal remain blocked.

Reports: `inventory_transactions` + integrity. Collections are not labelled profit.

Min/max/reorder remain on inventory items. No invented notification recipient list, VAT, or named variance-approver role.

Status: Implemented but not fully verified.

9. Roles and permissions matrix

Action	Permission	Extra rule
View stock / oversight	`inventory:read`	
Pharmacy POS / queue	`pharmacy:read`	Issue needs `pharmacy:dispense`
Receive, transfer, start/count/submit/reject stock-take	`inventory:manage`	
Approve and post stock-take	`inventory:manage` **or** `settings:manage`	Actor ≠ `submitted_by`
Collect payment	existing payments collect	
Documented credit	`payments:manage`	Not the preparer (`pharmacy-clearance`)
Hospital settings	`settings:manage`	Not granted to store users for Oversight

Enforced on controllers/services, not only by hiding buttons.

10. Database tables / migrations

Migration	Change	Why

`1772000000000-InventoryReservations`	`demo.inventory_reservations`	Holds without touching on-hand
`1772100000000-PharmacyWalkInCharges`	`patient_id` nullable on charges and payment_transactions	Walk-in pharmacy bills
`1772200000000-InventoryStockTakeSessions`	session + line tables	Count/approve without posting on count

Applied additively via `docker exec -i afya-sasa-postgres-1 psql` on **local** compose Postgres (`afya-sasa-postgres-1`, created 2026-09-21, database `afyasasa`, user `afyasasa`). Not production Contabo. Compose `.env` `TYPEORM_MIGRATIONS_RUN=false` was not flipped.

Rollback (local only): `DROP TABLE demo.inventory_stock_take_lines; DROP TABLE demo.inventory_stock_take_sessions;` Reservations/nullable patient_id rollback would restore NOT NULL only after no walk-in rows exist.

Isolated disposable `inventory_it` database: **not created** (credentials/host would be guessed). Migration tests against a throwaway DB: **Blocked**.

11. Reused components and removals

Reused: `inventory_items/batches/transactions`, payments charges, FEFO allocate, `adjustOnHand`, PaymentDesk, print receipt, HCC shell.

Removed: none. Incomplete CSV stock-take was **not** deleted; it is labelled legacy sheet import.

12. API routes and frontend screens changed

New/changed APIs:

`GET /inventory/oversight-summary`

`GET|POST /inventory/stock-take-sessions`

`POST .../lines/:lineId`

`POST .../submit|reject|approve`

Existing: `sales/prepare`, `sales/otc/prepare`, `sales/cancel`, `payments/pharmacy-queue`, `payments/pharmacy-sale/:saleId`.

Frontend: `PharmacyPosPanel`, `StockTakeSessionPanel`, `StockOversightPanels`, PaymentDesk pharmacy lookup, ErrorBoundary.

13. Tests run (actual commands)

```
npm --prefix backend test -- --runInBand --forceExit
```

Result: 87 suites passed, 469 passed, 20 skipped, 0 failed. Time ~8.6s. Open-handles warning only.

```
npm --prefix backend run build
npm --prefix frontend run build
```

Result: both succeeded (`tsc -b && vite build`). Frontend bundle in image: `index-CddTJreG.js` contains `Stock-take session` (count 1). Backend dist contains `stock-take-sessions`.

```
npm --prefix backend test -- --testPathPatterns=inventory-pg.integration
```

Not run — `INVENTORY_TEST_DATABASE_URL` / `inventory_it` not established.

14. Browser scenarios actually executed

Environment: Cursor browser tab `http://localhost:8080/` after `docker compose up -d --build frontend backend`. Home HTTP 200. Health HTTP 200.

#	Scenario	Result
1	Login and navigation per role	**Blocked.** Login form shown. `it@jalaram.co.ke` / `ChangeMe123!` → “Authentication is required.” Live staff password was not reset.
2–15	Catalogue, prepare, pay, issue, unpaid block, walk-in, receipt, retries, cancel, stock-take variants, expiry block, unauthorised approve, reports, empty/error	**Not executed.** Would claim false completion if inferred from unit tests.

Unauthenticated API checks from the sandbox could not bind to `127.0.0.1:8080`; the same host returned 200 from the browser and from an unsandboxed curl to `/` and `/api/v1/health`.

15. Known limitations and unfinished work

- Browser E2E with live credentials.
- Named variance-approver role (segregation is same-permission, different user).
- Recall / quarantine / disposal / customer returns.
- Insurance, waiver, POS discount, POS tax.
- Automatic reservation expiry.
- Isolated migration/pg integration harness.
- CSV confirm-and-post still bypasses independent approval if `inventory:manage` uses it.
- `.env` `POSTGRES_HOST=localhost` `POSTGRES_PORT=5432` is not the compose mapped `5433`; compose backend uses internal `postgres:5432`. Do not point host tools at the wrong instance.

16. Data integrity, migration and recovery

Reservations do not enter `inventory_transactions`. Integrity reports still reconcile on-hand to the ledger. Stock-take post is transactional and idempotent by session id. Counting cannot change stock. Concurrent dispense during an open count fails post on stale snapshot.

Backups: existing `ops/backup-postgres.sh`. Destructive `docker compose down -v` and `dev:fresh` were not used.

17. Files changed and decisions

Uncommitted working tree (do not commit until asked). New this completion pass: stock-take session entities/migration/service/UI, oversight summary, session tests, this report.

Decisions: one ledger; walk-in without dummy patient; independent approval without a new permission key; fail-closed insurance; Stock Oversight on `inventory:read`.

18. Release-readiness checklist and next steps

Check	Status
Local compose identity preserved	Yes
`TYPEORM_MIGRATIONS_RUN=false`	Yes

No production touch / no commit	Yes
Automated unit suite green	Yes (469 pass / 20 skip)
Frontend production build	Yes
Images rebuilt	Yes
15 browser scenarios	No
Isolated migration test	No
Policy (insurance, recall, tax)	Fail closed / blocked

Recommended next steps (user): sign in with the live account already used in this hospital (`userId` `20000000-0000-4000-8000-000000000001` was active on nginx logs). Walk Pharmacy POS, cashier queue, issue, and Stock take with a second inventory user for approve. Only then consider a commit.

Requirement labels

Area	Label
Protect existing app / identity	Implemented and verified
Rx prepare → reserve → pay → issue	Implemented but not fully verified
POS multi-line + walk-in	Implemented but not fully verified
Cashier clearance fail-closed	Implemented but not fully verified (unit)
Control Center four areas	Implemented but not fully verified
Stock-take session lifecycle	Implemented but not fully verified
Quarantine / recall / disposal	Blocked
POS discount / tax / refunds	Not implemented (policy)
Isolated DB migration tests	Blocked
15 browser E2E scenarios	Blocked (login)
Management PDF	Implemented if generated beside this file

Controlled release: no. Local verification build: **yes**, pending live login.